Math 421/510: Homework 2

1. Let ℓ_c denote the space of finite sequences, endowed with the supremum norm $\|\cdot\|_\infty$. Define $T : \ell_c \to \ell_c$ by

$$T : (x_1, x_2, x_3, \ldots) \mapsto \left(x_1, \frac{1}{2}x_2, \frac{1}{3}x_3, \ldots\right).$$

Show that T is well-defined, linear, bounded, and bijective, but that T^{-1} is not bounded. Why does this not contradict the open mapping theorem?

Solution.

First, we confirm T is well defined. For $x \in \ell_c$, Tx is a well defined sequence by definition, so it remains to check if $Tx \in \ell_c$. Since x is a finite sequence, there exists N such that $x_n = 0$ for $n > N$. However, since $(Tx)_n = \frac{1}{n}x_n$, $x_n = 0 \Rightarrow (Tx)_n = 0$ so then Tx is also a finite sequence, so $Tx \in \ell_c$.

For linearity, we see that

$$T(x + \lambda y) = (x_1 + \lambda y_1, \frac{1}{2}x_2 + \frac{\lambda}{2}y_2, \ldots) = (x_1, \frac{1}{2}x_2, \ldots) + \lambda(y_1, \frac{1}{2}y_2, \ldots) = Tx + \lambda Ty.$$

Therefore, T is linear.

To show boundedness, consider $\|x\|_\infty = 1$. Then, $|(Tx)_n| = \frac{1}{n}|x_n| \le \frac{1}{n} \le 1$. Thus, $\|Tx\|_\infty = \sup_n |(Tx)_n| \le 1$, and since this applies for all x with $\|x\|_\infty = 1$ it follows that $\|T\| \le 1$.

To show T is a bijection, we show injectivity and surjectivity. Suppose $Tx = 0$. Then, $\frac{1}{n}x_n = 0 \Rightarrow x_n = 0 \forall n \in \mathbb{N} \Rightarrow x = 0$. Now, suppose $\{y_n\} \in \ell_c$. Then, if we define $\{x_n\}$ such that $x_n = ny_n$, we have $(Tx)_n = \frac{1}{n}ny_n = y_n$, so $Tx = y$. This also gives us that $T^{-1}(x_1, x_2, x_3, \ldots) = (x_1, 2x_2, 3x_3, \ldots)$.

Finally, we show that T^{-1} is not bounded. Consider a sequence of n 1s, that is, $x = (1, \ldots, 1, 0, \ldots)$. Then, $T^{-1}(x) = (1, 2, 3, \ldots, n, 0, 0, \ldots)$. But then, $\|T^{-1}(x)\|_\infty = n$. Therefore, for all n, there exists some $x \in \ell_c$ with $\|x\|_\infty = 1$ such that $\|T^{-1}x\|_\infty = n$, so T^{-1} is unbounded.

This does not contradict the Corollary to the open mapping theorem, because ℓ_c is not a banach space (that is, it is not complete) w.r.t. the supremum norm. To see this, consider that the sequence $y^{(n)} = (1, \ldots, \frac{1}{n}, 0, \ldots)$ is Cauchy (indeed, if w.l.o.g. $m > n$, then $\|y^{(n)} - y^{(m)}\| = \sup_i |y_i^{(n)} - y_i^{(m)}| = 1/(n+1)$), but its limit is an infinite sequence, so $\lim_{n\to\infty} y^{(n)} \notin \ell_c$. Therefore, the requirements of the Open Mapping Theorem that $\mathcal{X}$ and $\mathcal{Y}$ are Banach spaces is not fulfilled, so we can't expect the theorem to hold.

2. Let $Y \subset \ell^1$ denote the space of sequences $x = (x_1, x_2, \ldots)$ for which $\sum_{n=1}^\infty n|x_n| < \infty$, endowed with the norm $\|x\|_1 = \sum_{n=1}^\infty |x_n|$. Define the linear map $T : Y \to \ell^1$ by

$$(Tx)_n = nx_n.$$

Show that T is closed, but not bounded. Why does this not contradict the closed graph theorem?

Solution.

We show that T is closed. Suppose that $x^{(n)} \in Y$ is a sequence such that $x^{(n)} \to x \in Y$, and suppose that we know that $Tx^{(n)} \to y \in \ell^1$. For T to be closed, we need to show that $Tx = y$. Fix $m \in \mathbb{N}$. Then, we have that

$$\begin{aligned} |(Tx)_m - (y)_m| &\leq |(Tx)_m - (Tx^{(n)})_m| + |(Tx^{(n)})_m - y_m| \\ &\leq m|(x)_m - (x^{(n)})_m| + \|Tx^{(n)} - y\|_1 \\ &\leq m\|x - x^{(n)}\|_1 + \|Tx^{(n)} - y\|_1 \\ &\to 0 \text{ as } n \to \infty. \end{aligned}$$

Therefore, $(Tx)_m = y_m$ for all m, so $Tx = y$. Thus, T is closed.

However, T is not bounded. Consider $x^{(n)} = (0, \ldots, 0, 1, 0, \ldots)$, where the 1 is in the n-th location. Then, $\|x^{(n)}\|_1 = 1$, and $\sum_{m=1}^{\infty} m|(x^{(n)})_m| = n < \infty$, so $x^{(n)} \in Y$. However, $\|Tx^{(n)}\|_1 = n$. Therefore, $\|T\| \geq n$ for all $n \in \mathbb{N}$, so we have that T is not bounded.

This does not contradict the closed graph theorem, because Y is not a Banach space. To show, this, we construct a sequence in Y which is cauchy that converges to an element in $\ell^1 \setminus Y$. Consider the sequence $x_n = \frac{1}{n^2}$, and let $x^{(N)}$ be the sequence such that $x_n^{(N)} = 0$ if $n > N$, and $x_n^{(N)} = (x)_n$ if $n \leq N$. Then, since $x^{(N)}$ is a finite sequence, $x^{(N)} \in Y$. However, $\lim_{N\to\infty} x^{(N)} = x$ under the $\|\cdot\|_1$ norm, since for $N > M$, we have

$$\|x - x^{(N)}\| = \sum_{n=N}^{\infty} \frac{1}{n^2} < \sum_{n=M}^{\infty} \frac{1}{n^2} \to 0 \text{ as } M \to \infty.$$

Also, $\|x\| = \sum_{n=1}^{\infty} \frac{1}{n^2} = \pi^2/6$, so $x \in \ell^1$. However, $\sum_{n=1}^{\infty} n|(x)_n| = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$, so $x \notin Y$. Therefore, we have a sequence in Y, $\{x^{(N)}\}_{N=1}^{\infty}$ which converges to x, but $x \notin Y$. Therefore, Y is not closed, so it is not complete, so it is not Banach. Even if $T : Y \to \ell^1$ is closed, the hypothesis of the closed graph theorem that both Y and ℓ^1 are banach is not satisfied.

3. ([Folland, ex. 5.39]): Let X, Y and Z be Banach spaces, and $B : X \times Y \to Z$ be such that $\forall\, x \in X$, $B(x, \cdot) : Y \to Z \in L(Y, Z)$ and $\forall\, y \in Y$, $B(\cdot, y) : X \to Z \in L(X, Z)$.
   (a) show $\|B(x, y)\| \leq C\|x\|\|y\|$ for some C;
   (b) show that B is continuous (as a map from Banach space $X \times Y$ to Z.

Solution.

(a) Fix $x \in X$, with $x \neq 0$. Consider the linear map $\frac{B(x,\cdot)}{\|x\|} : \mathcal{Y} \to \mathcal{Z}$. Then, since $B(x, \cdot) \in L(Y, Z)$, there exists some constant C_x such that $\|B(x, \cdot)(y)\| \leq C_x\|y\|$, and so $\|\frac{B(x,\cdot)(y)}{\|x\|}\| \leq \frac{C_x}{\|x\|}\|y\|$.

Now, fix $y \in Y$. Since $B(\cdot, y) \in L(X, Z)$, we have that $\|B(x, y)\| \leq \|B(\cdot, y)\|\|x\|$. However, this implies that $\frac{B(x,y)}{\|x\|} \leq \|B(\cdot, y)\|$ for all $x \in X$ with $x \neq 0$, so

$$\sup_{x\in X\setminus\{0\}} \|B(x, \cdot)(y)/\|x\|\| \leq \|B(\cdot, y)\|.$$

Since Y is a Banach space, and this holds for all $y \in Y$, the requirements of the Uniform Boundedness Principle are satisfied, so we have that $\sup_{x\in X\setminus\{0\}} \|\frac{B(x,\cdot)}{\|x\|}\| \leq M$ for some $M \in \mathbb{R}$. Then, $\|B(x, \cdot)\| \leq M\|x\|$, which by definition means that $\|B(x, y)\| \leq M\|x\|\|y\|$ when $\|x\| \neq 0$. When, $x = 0$, then clearly $\|B(0, \cdot)\| = 0 \leq M$.

(b) Let $(x,y) \in X \times Y$. Then, from part (a), we have some M such that $\|B(x,y)\| \leq M\|x\|\|y\|$. Let $\varepsilon > 0$. Choose $\delta > 0$ such that $\delta < \min\{\frac{\varepsilon}{3M\|x\|}, \frac{\varepsilon}{3M\|y\|}, \frac{\sqrt{\varepsilon}}{\sqrt{3M}}\}$. Then, when $\|(\Delta x, \Delta y)\| < \delta$ so that $\|\Delta x\|, \|\Delta y\| < \delta$, we have

$$\begin{aligned}\|B(x+\Delta x, y+\Delta y) - B(x,y)\| &= \|B(x,y) + B(\Delta x, y) + B(x, \Delta y) + B(\Delta x, \Delta y) - B(x,y)\| \\ &=\leq \|B(\Delta x, y)\| + \|B(x, \Delta y)\| + \|B(\Delta x, \Delta y)\| \\ &\leq M\|\Delta x\|\|y\| + M\|x\|\|\Delta y\| + M\|\Delta x\|\|\Delta y\| \\ &< M\delta\|y\| + M\|x\|\delta + M\delta^2 \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.\end{aligned}$$

Therefore, $B(x,y)$ is continuous as a bilinear map from $X \times Y$ to Z.

4. Suppose $\{g_n\}_{n=1}^{\infty} \subset L^{\infty}$ with $\|g_n\|_{\infty} \leq C$, and $g_n \to g$ a.e. Show $g \in L^{\infty}$ with $\|g\|_{\infty} \leq C$.

Solution.

We need to show that $g : X \to \mathbb{R}$ is measurable, and $\|g\|_{\infty} < \infty$. By proposition 2.11 b. in Folland, since g_n is measurable for $n \in \mathbb{N}$, and $g_n \to g$ a.e., g is measurable. Now, for $D > C$, we have that

$$\begin{aligned}\{x : |g(x)| > D\} &\subset \{x : g(x) \neq \lim_{n\to\infty} g_n(x)\} \cup \{x : \limsup |g_n(x)| > D\} \\ &= \{x : g(x) \neq \lim_{n\to\infty} g_n(x)\} \cup \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{x : |g_n(x)| > D\} \\ &\subset \{x : g(x) \neq \lim_{n\to\infty} g_n(x)\} \cup \bigcup_{n=1}^{\infty} \{x : |g_n(x)| > D\}.\end{aligned}$$

But then,

$$\mu(\{x : |g(x)| > D\}) \leq \mu(\{x : g(x) \neq \lim_{n\to\infty} g_n(x)\}) + \sum_{n=1}^{\infty} \mu(\{x : |g_n(x)| > D\}) = 0 + \sum_{n=1}^{\infty} 0.$$

Since this applies for all $D > C$, we have that

$$\inf\{a \geq 0 : \mu(\{x : |g(x)| > a\}) = 0\} \leq C,$$

as desired. In particular, $\|g\|_{\infty} \leq C$, and $\|g\|_{\infty} < \infty$, so $g \in L^{\infty}$.

5. This exercise gives an another example of a situation where the linear isometry $L^1 \to (L^{\infty})^*$ is not surjective. In assignment #1, you showed the existence (by invoking the H-B theorem) of $T \in (\ell^{\infty}, \|\cdot\|_{\infty})^*$, with $\|T\| = 1$, and $TS = S$ where $S : (x_1, x_2, x_3, \ldots) \mapsto (x_2, x_3, \ldots)$ is the shift operator. Show that T is not given by $Tx = y \cdot x$ for any $y \in \ell^1$.

Solution.

We showed last homework in problem 6 (e) that if $x = (x_1, x_2, \ldots) \in \ell^{\infty}$ satisfies $\lim_{k\to\infty} x_k = L$, then $Tx = L$. Let $y \in \ell^1$ be arbitrary. In particular, $\sum_{n=1}^{\infty} |y_n| < \infty$. Then, there exists some N such that $\sum_{n=N}^{\infty} |y_n| < 1$. Now, consider $x = (x_1, x_2, \ldots)$ such that $x_i = 0$ for $i < N$, and $x_i = 1$ for $i \geq N$. In particular, $\lim_{i\to\infty} x_i = 1$, so by 6 (e) from last homework, we have $Tx = 1$. However,

$$|y \cdot x| = \left|\sum_{n=1}^{\infty} x_n y_n\right| = \left|\sum_{n=N}^{\infty} y_n\right| \leq \sum_{n=N}^{\infty} |y_n| < 1,$$

so $y \cdot x \neq 1 = Tx$. Since $y \in \ell^1$ was arbitrary, T is not given by $y \cdot x$ for any $y \in \ell^1$.